

ApproxDA-TransUNet: Understanding Global Context Sensitivity of Attention Approximation for Medical Image Segmentation

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Abstract—Dual-attention mechanisms have demonstrated strong performance in medical image segmentation by jointly modeling long-range spatial and channel dependencies, yet their quadratic complexity has motivated numerous approximation strategies whose task-dependent behavior remains poorly understood. To investigate this, we propose Approximate Dual Attention TransUNet (ApproxDA-TransUNet), a dual-attention approximation framework for hybrid CNN-Transformer architectures that integrates controllable windowed low-rank spatial attention with grouped channel attention. Beyond serving as an efficiency mechanism, ApproxDA-TransUNet provides a parametric platform to systematically evaluate how attention approximation interacts with representation capacity across diverse clinical segmentation tasks. On Synapse Multi-Organ CT, Kvasir-SEG polyp, and ISIC 2018 skin lesion benchmarks, ApproxDA-TransUNet achieves 80.94%, 90.17%, and 89.58% Dice, improving Dice over DA-TransUNet across all three datasets. Furthermore, our empirical analysis reveals two key insights. First, window sensitivity varies markedly across tasks: the Dice spread across window sizes is up to $4.6\times$ larger on multi-organ CT than on polyp and lesion targets, a characteristic we formalize as empirical Global Context Sensitivity (GCS). Second, in our multi-organ CT experiments, the learnable routing gate converged near uniform blending without outperforming single-branch spatial attention. These findings demonstrate that attention approximation functions not merely as a computational shortcut, but as a scale-dependent inductive bias whose optimal configuration is governed by task-specific context requirements.

Index Terms—medical image segmentation, dual attention, attention approximation, window size, global context sensitivity

I. INTRODUCTION

Transformers have advanced medical image segmentation by enabling global long-range dependency modeling that convolutional networks capture only indirectly. Dual attention [1]—combining a Position Attention Module (PAM) for spatial correlations and a Channel Attention Module (CAM) for channel correlations—has emerged as a powerful mechanism, and methods such as DA-TransUNet [2] demonstrated its effectiveness in hybrid CNN-Transformer architectures for multi-organ and polyp segmentation. The quadratic complexity of dual attention, however, has prompted a wave of efficient approximations—windowed attention [3], low-rank factorization [4], and grouped channel interaction [5]—that are now

standard design choices. These strategies are usually adopted under the implicit assumption that approximation is broadly applicable with minor accuracy trade-offs—an assumption rarely examined across tasks with different spatial characteristics:

When, and at what scale, is attention approximation beneficial for medical image segmentation?

To investigate this question, we design **ApproxDA-TransUNet**¹: a dual-attention architecture featuring three independently controllable approximation axes—windowed PAM, low-rank projection, and grouped CAM. ApproxDA-TransUNet provides both competitive segmentation performance across benchmarks and a controllable framework for studying the spatial and channel granularity of attention.

Applying ApproxDA-TransUNet across three medical segmentation benchmarks reveals distinct task-dependent characteristics: the windowed attention operator is highly window-sensitive on multi-organ CT yet largely robust on polyp and skin lesion segmentation. With an appropriately selected window, ApproxDA-TransUNet improves Dice on all three benchmarks over DA-TransUNet, while demonstrating that context requirements diverge fundamentally across clinical targets. We further observe that, on multi-organ CT, learnable routing between spatial and channel branches converges to near-uniform weighting without surpassing single-branch PAM routing.

The main contributions of this paper are:

- **ApproxDA-TransUNet Framework:** We propose ApproxDA-TransUNet, a dual-attention approximation architecture featuring three independently parameterizable operators—windowed PAM, low-rank projection, and grouped CAM—achieving consistent Dice gains over DA-TransUNet across three diverse benchmarks.
- **Systematic Context Sensitivity Study:** Using ApproxDA-TransUNet as a controlled diagnostic platform, we present a systematic empirical study of

¹Source code will be made available upon acceptance.

attention approximation across heterogeneous medical segmentation tasks, observing up to a $4.6\times$ difference in window sensitivity across benchmarks. We formalize this behavior via an empirical *Global Context Sensitivity* (GCS) metric, quantifying the architectural response to window scale across distinct anatomical targets.

- **Empirical Insights on Routing and Inductive Bias:** Our findings demonstrate that attention approximation can serve as a beneficial locality-induced bias for appearance-dominated segmentation, and document that learnable dual-branch routing tends toward static uniform fusion on multi-organ CT.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA-TransUNet framework; Section IV compares it with published methods; Section V reports ablation studies and analysis; Section VI concludes.

II. RELATED WORK

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [6] introducing the skip-connection design that underpins most segmentation networks, and successors [7]–[10] progressively adding dense connections, attention gating, and residual learning for improved feature reuse and training stability. These models offer strong computational efficiency and favorable inductive biases for local texture patterns, but their fixed receptive fields cannot capture long-range spatial dependencies across anatomical structures, limiting performance on tasks requiring global context.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-UNet [12], UNETR [13], UCTransNet [14], TransNorm [15], and DAEformer [16] combine convolutional feature extraction with transformer-based contextual modeling and report strong segmentation performance across a wide range of medical imaging tasks. These methods treat global context modeling as uniformly beneficial, however, without examining whether different tasks require different degrees of spatial context.

C. Dual Attention and DA-TransUNet

DANet [1] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, improving accuracy while inheriting the quadratic cost of full-map attention. DBAANet [17] further extends dual-branch attention by aggregating position and channel attention across multiple scales, demonstrating the continued

appeal of dual-attention paradigms for medical image segmentation.

D. Efficient Attention and Adaptive Routing

Windowed local attention [3], low-rank projection [4], and grouped channel interaction [5] have become standard strategies for reducing the computational footprint of global attention, with learnable gating widely adopted to blend multi-branch representations [5], [16]. DAMamba [18] further demonstrates that dual-attention routing transfers effectively to state-space models for lightweight breast ultrasound segmentation, highlighting the versatility of multi-branch attention paradigms. While existing efficient attention methods are predominantly evaluated by computational metrics, their task-dependent impact on feature representation and segmentation accuracy remains under-explored. In this work, we systematically investigate this interaction by varying approximation parameters across diverse clinical tasks and evaluating the empirical behavior of adaptive branch routing.

III. METHOD

This section introduces the overall architecture of ApproxDA-TransUNet and the placement of ApproxDABlocks within it, details the three independently controllable approximation axes implemented in each block, and analyzes their computational characteristics.

A. Architecture Overview

Figure 1 illustrates the overall architecture: ApproxDA-TransUNet adopts the hybrid CNN-ViT encoder-decoder structure of DA-TransUNet [2], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce features at 112×112 , 56×56 , and 28×28 resolutions (strides 2, 4, and 8 for a 224×224 input), while a 12-layer ViT processes 14×14 patch tokens at the bottleneck. While DA-TransUNet integrates attention modules at three stages (the bottleneck and the 512- and 256-channel skip paths), ApproxDA-TransUNet uniformly equips all three skip connections (512 channels at 28×28 , 256 at 56×56 , and 64 at 112×112) as well as the bottleneck (768 channels at 14×14) with ApproxDABlocks, providing comprehensive multi-scale context modulation. In contrast to channel-compression designs that project intermediate features to $C/16$, each ApproxDABlock operates on the full C channel capacity during token projection to preserve representational fidelity. Each block routes activations through a Low-Rank Windowed PAM branch and a Grouped CAM branch, fused via a configurable gating vector g . Figure 2 details the internal structure of the ApproxDABlock and its three parameterized approximation axes.

B. Attention Approximation Formulation

Standard dual attention performs dense spatial and channel affinity modeling over the entire feature map, incurring quadratic complexity in both spatial resolution and channel count. ApproxDA-TransUNet structures this complexity along

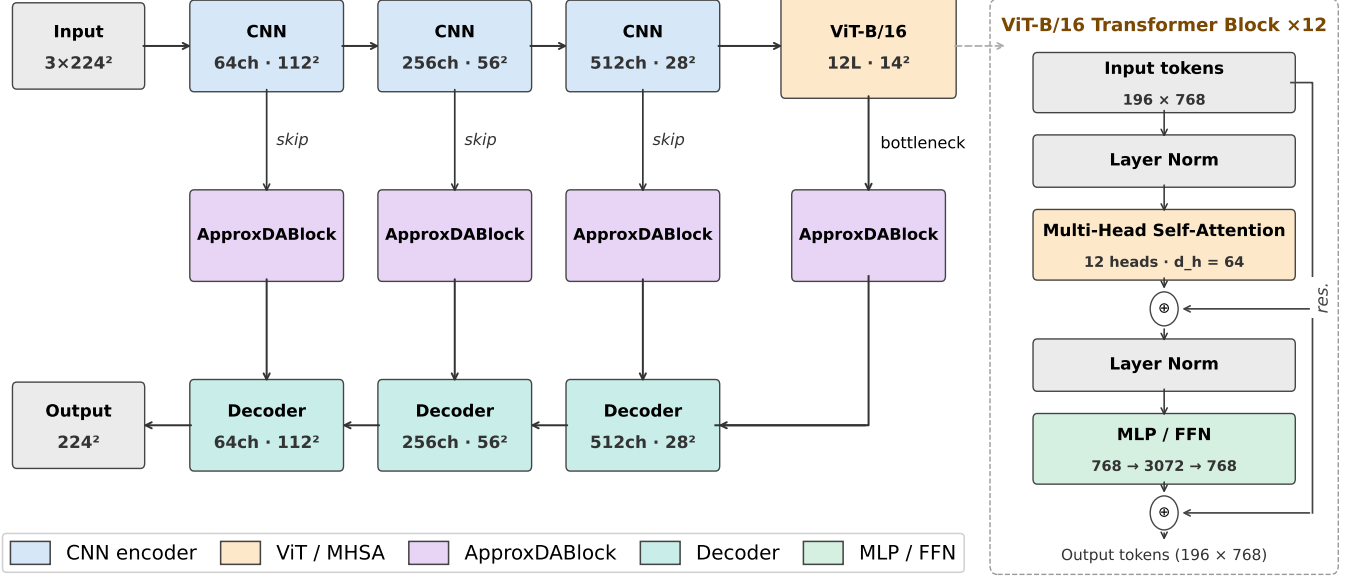


Fig. 1. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) are placed at all three skip connections and at the bottleneck. *gate_mode* controls branch routing during ablation.

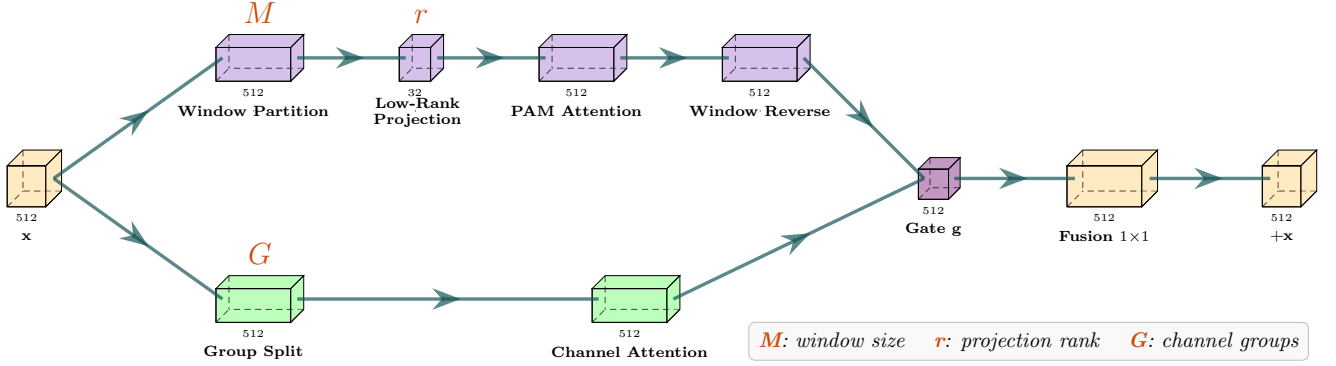


Fig. 2. ApproxDABlock internals with three independently controllable approximation axes: spatial scope (M , window size), representation fidelity (r , projection rank), and channel interaction (G , group count). Input $\mathbf{x}$ splits into a Low-Rank Windowed PAM branch (lavender, upper) and a Grouped CAM branch (green, lower); the per-channel gate g fuses the outputs, followed by a 1×1 convolution, and $\mathbf{x}$ is added as a residual.

three orthogonal axes, each governing a distinct dimension of the approximation:

Spatial scope (window size M) controls the receptive range of spatial attention. Full global attention ($M=H$) models all pairwise spatial dependencies, whereas local $M \times M$ window partitioning restricts attention to local neighborhoods at reduced complexity.

Representation fidelity (rank r) controls the subspace precision within each window. Low-rank projection ($r \ll M^2$) compresses the key-value affinity matrix into an informative subspace, while higher r approaches exact intra-window attention.

Channel interaction (groups G) controls the cross-channel dependency span in CAM. Partitioning C channels into G

independent groups constrains channel correlations to within-group interactions, scaling the interaction granularity.

Because these parameters are orthogonally parameterized, any element in $\{M, r, G\}$ can be independently ablated while maintaining the others constant.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA-TransUNet partitions the feature map into non-overlapping windows of size $M \times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$, where $N = HW$.

Window partitioning. Following [3], we partition the $C \times H \times W$ feature map into n_W non-overlapping $M \times M$

windows, with $M \leftarrow \min(M, H, W)$. For one window, let $\mathbf{X} \in \mathbb{R}^{C \times N_w}$ with $N_w = M^2$ denote its features, and let

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{X}, \quad \mathbf{K} = \mathbf{W}_K \mathbf{X}, \quad \mathbf{V} = \mathbf{W}_V \mathbf{X} \in \mathbb{R}^{C \times N_w} \quad (1)$$

be the query, key, and value features produced by 1×1 convolutions $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{C \times C}$ (with bias).

Low-rank projection. Following [4], keys and values inside each window are compressed along the token axis by a learned projection $\mathbf{P} \in \mathbb{R}^{N_w \times r}$, shared by keys and values and across windows within a block:

$$\tilde{\mathbf{K}} = \mathbf{K} \mathbf{P}, \quad \tilde{\mathbf{V}} = \mathbf{V} \mathbf{P} \in \mathbb{R}^{C \times r}. \quad (2)$$

Each of the N_w queries attends to the r projected keys:

$$\begin{aligned} \mathbf{A} &= \text{softmax}(\mathbf{Q}^\top \tilde{\mathbf{K}}) \in \mathbb{R}^{N_w \times r}, \\ \mathbf{E} &= \alpha (\mathbf{A} \tilde{\mathbf{V}}^\top)^\top + \mathbf{X} \in \mathbb{R}^{C \times N_w}, \end{aligned} \quad (3)$$

where the softmax normalizes each row across the r projected keys, without temperature scaling (consistent with standard PAM [1]), and α is a learned scalar initialized to 0. Each window output $\mathbf{E}$ is reshaped to $C \times M \times M$, and all windows are reassembled into the $H \times W$ grid. The full $N_w \times N_w$ affinity is thus substituted by an $N_w \times r$ matrix, lowering attention complexity to $\mathcal{O}(NrC)$. When M is clamped to match feature dimensions, the leading N_w rows of $\mathbf{P}$ are utilized; setting $M=112$ establishes global spatial scope per feature map while preserving low-rank ($r=32$) projection. Together, windowing and low-rank projection reduce the theoretical attention-matrix complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(Nr)$ ($\sim 390 \times$ reduction in affinity matrix entries at 112×112 with $r=32$; see Section IV-C).

D. Channel Approximation

The standard Channel Attention Module computes a full $C \times C$ channel correlation matrix at $\mathcal{O}(C^2 N)$ complexity. ApproxDA-TransUNet approximates this by partitioning C channels into G groups of width $C_g = C/G$ and computing independent per-group attention. For group g with channel slices $\mathbf{A}_1^{(g)}, \dots, \mathbf{A}_{C_g}^{(g)} \in \mathbb{R}^N$ of the block input,

$$\begin{aligned} x_{ij}^{(g)} &= \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_k^{(g)})}, \\ \mathbf{E}_j^{(g)} &= \beta \sum_{i=1}^{C_g} x_{ij}^{(g)} \mathbf{A}_i^{(g)} + \mathbf{A}_j^{(g)}, \end{aligned} \quad (4)$$

with a learned scalar β initialized to 0 and total cost $\mathcal{O}(C^2 N/G)$. In Eq. (4), softmax normalizes along keys to yield row-stochastic attention weights before aggregation. In our subsequent primary benchmarks, PAM-only routing is utilized, focusing analysis on spatial context modulation.

E. Routing Strategy

The routing mechanism provides a configurable `gate_mode` to blend spatial and channel representations. PAM and CAM branch outputs $\mathbf{P}$ and $\mathbf{C}$ (both $C \times H \times W$) are fused via:

$$\mathbf{y} = \mathbf{W}_f (\mathbf{g} \odot \mathbf{P} + (\mathbf{1} - \mathbf{g}) \odot \mathbf{C}) + \mathbf{x}, \quad (5)$$

where $\mathbf{W}_f$ is a 1×1 convolution and $\mathbf{g} \in [0, 1]^C$ is a per-channel routing vector broadcast spatially, determined by the `gate_mode` setting:

$$\begin{aligned} \text{pam:} & \quad \mathbf{g} = \mathbf{1} \quad (\text{PAM only}) \\ \text{cam:} & \quad \mathbf{g} = \mathbf{0} \quad (\text{CAM only}) \\ \text{fixed:} & \quad \mathbf{g} = 0.5 \cdot \mathbf{1} \quad (\text{equal blend}) \\ \text{learn:} & \quad \mathbf{g} = \sigma(\mathbf{W}_g \text{GAP}(\mathbf{x})) \in (0, 1)^C \end{aligned}$$

In `learn` mode, $\mathbf{g}$ is derived per sample from globally pooled features (GAP) via linear projection $\mathbf{W}_g$ and sigmoid activation σ . This formulation establishes a unified parametric interface to systematically ablate branch contributions without altering backbone topology.

The network is trained with a composite Dice and cross-entropy objective with equal weighting:

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{Dice}} + \frac{1}{2} \mathcal{L}_{\text{CE}}, \quad (6)$$

maintaining standard practice for medical segmentation benchmarks [2], [11].

IV. RESULTS

We evaluate ApproxDA-TransUNet on three benchmarks spanning tasks with different spatial characteristics, comparing against DA-TransUNet and other published baselines.

A. Experimental Setup

Datasets. We evaluate on three representative medical image segmentation benchmarks.

Synapse Multi-Organ CT [19]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes. Metrics: mean DSC and HD95.

Kvasir-SEG [20]: 1000 endoscopy polyp images (binary) with an 800/200 train/test split. Metrics: DSC and mIoU.

ISIC 2018 [21]: 2594 dermoscopy skin lesion images (binary) with a 2075/519 train/test split. Metrics: DSC and mIoU.

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K, trained on NVIDIA Tesla T4 (16GB) GPUs. Optimizer: SGD, lr=0.01 with polynomial decay; batch size 24; input 224×224 ; 300 epochs; random rotation and flipping. Default ApproxDA-TransUNet configuration: $M=7$, $r=32$, $G=8$.

Evaluation Protocol. In accordance with established benchmark evaluation protocols for Synapse and related benchmarks [2], [11], models are trained on the training partition and evaluated periodically (every 15 epochs) on the test split to identify peak checkpoint performance. Hyperparameter configurations (M, r, G , and gating mode) reported in Tables I–III correspond to the optimal settings identified in the comprehensive ablation sweeps in Section V. In line with standard single-split reporting in this benchmark literature, results reflect single training runs.

Gate Protocol. ApproxDA-TransUNet supports four gate modes: `pam` ($\mathbf{g}=\mathbf{1}$, PAM only), `cam` ($\mathbf{g}=\mathbf{0}$, CAM only), `fixed` ($\mathbf{g}=0.5 \cdot \mathbf{1}$, equal blend), and `learn` ($\mathbf{g}$ learned per channel). Each experiment’s gate configuration is specified in the corresponding table footnote.

B. Comparison with Published Methods

Tables I–III compare ApproxDA-TransUNet against published baselines across all three benchmarks. Baseline values are compiled from published literature: for Synapse from the respective references compiled in [2], and for Kvasir-SEG and ISIC 2018 from Table 2 of DA-TransUNet [2] (256×256 input, Adam, random 75/25 split). These literature metrics provide indicative benchmarks across architectures.

Synapse. Table I benchmarks against 11 published methods. CNN-based models plateau at 76–78% DSC, with AttUNet leading at 77.77%. Transformer-based methods progressively improve: TransUNet (77.48%), UCTransNet (78.23%), MIM (78.59%), Swin-Unet (79.13%), and DA-TransUNet (79.80%). ApproxDA-TransUNet with $\text{gate}=\text{pam}$, $M=28$, $r=32$ achieves **80.94%** DSC, outperforming all compared baselines and improving by +1.14% over DA-TransUNet. Its HD95 reaches 27.49 mm (compared to 21.55 mm for Swin-Unet and 23.48 mm for DA-TransUNet). Section V examines the underlying window-size dynamics.

TABLE I SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

	Method	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
CNN-based	U-Net [6]	76.85	39.70
	U-Net++ [7]	76.91	36.93
	Residual U-Net [9]	76.95	38.44
	MultiResUNet [10]	77.42	36.84
	Att-Unet [8]	77.77	36.02
Transformer-based	TransUNet [11]	77.48	31.69
	UCTransNet [14]	78.23	26.75
	TransNorm [15]	78.40	30.25
	MIM [22]	78.59	26.59
	Swin-Unet [12]	79.13	21.55
	DA-TransUNet [2]	<u>79.80</u>	<u>23.48</u>
	ApproxDA-TransUNet (ours)	80.94	27.49

ApproxDA: $\text{gate}=\text{pam}$, $M=28$, $r=32$, $G=8$.

Kvasir-SEG. On polyp segmentation (Table II), U-Net leads the CNN-based group (88.22% DSC) while TransUNet (87.91%) and DA-TransUNet[†] (88.44%) are the strongest Transformer-based entries. ApproxDA-TransUNet with $\text{gate}=\text{pam}$, $M=56$ achieves **90.17%** DSC and **84.27%** mIoU, attaining top performance with a +1.73% DSC improvement over matched re-run DA-TransUNet and lower boundary distance (44.35 vs. 53.04 pixels). HD95 on Kvasir-SEG is reported in pixels due to lack of physical spacing calibration.

ISIC 2018. On skin lesion segmentation (Table III), CNN-based methods cluster between 83–88% DSC, with TransUNet (88.78%) and DA-TransUNet (88.88%) leading among Transformer-based approaches. ApproxDA-TransUNet with $\text{gate}=\text{learn}$, $M=7$ achieves **89.58%** DSC (+0.70% over reported DA-TransUNet), while mIoU (82.66%) is comparable to DA-TransUNet (82.78%). Baseline values are referenced

TABLE II SEGMENTATION RESULTS ON KVASIR-SEG POLYP

Method	DSC (%) $\uparrow$	mIoU (%) $\uparrow$	HD95 (px) $\downarrow$
U-Net [6]	88.22	80.12	–
Att-Unet [8]	86.61	78.01	–
U-Net++ [7]	86.57	77.67	–
Res-UNet [9]	77.85	66.04	–
TransUNet [11]	87.91	80.03	–
DA-TransUNet [2] [†]	<u>88.44</u>	<u>81.70</u>	<u>53.04</u>
ApproxDA-TransUNet (ours)	90.17	84.27	44.35

[†]DA-TransUNet re-run with our training settings and split (DSC 88.44%, vs. 88.47% reported in [2]); all other rows are literature values from [2]. ApproxDA: $\text{gate}=\text{pam}$, $M=56$, $r=32$.

from [2]. The minimal performance spread across window sizes on this dataset is detailed in Section V-B.

TABLE III SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

Method	DSC (%) $\uparrow$	mIoU (%) $\uparrow$
U-Net [6]	87.22	81.14
Att-Unet [8]	87.60	81.51
U-Net++ [7]	87.30	81.33
Res-UNet [9]	83.32	76.51
TransUNet [11]	88.78	82.63
DA-TransUNet [2]	<u>88.88</u>	<u>82.78</u>
ApproxDA-TransUNet (ours)	89.58	<u>82.66</u>

Baseline values from Table 2 of [2]. ApproxDA: $\text{gate}=\text{learn}$, $M=7$, $r=32$, $G=8$.

Overall, ApproxDA-TransUNet demonstrates consistently high DSC across all three benchmarks (Figure 3 illustrates qualitative segmentations), while showing that performance sensitivity to window scale is strongly task-dependent.

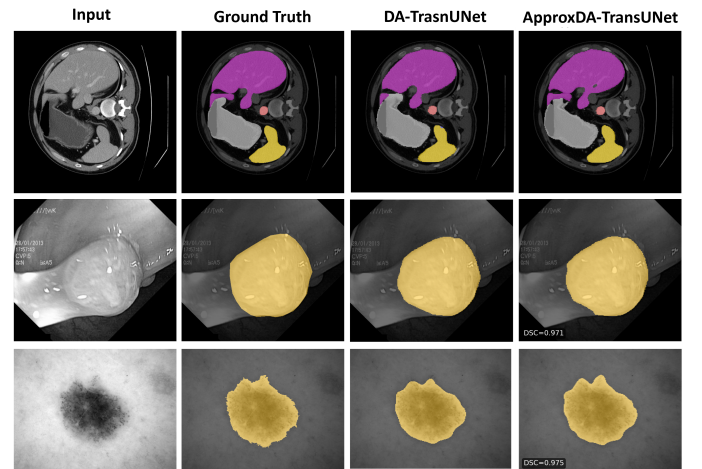


Fig. 3. Qualitative segmentation comparison (columns: input, ground truth, DA-TransUNet, ApproxDA). **Top (Synapse, $M=28$):** multi-organ CT; ApproxDA 80.94% DSC (+1.14%). **Middle (Kvasir-SEG, $M=56$):** polyp; 90.17% DSC (+1.73%). **Bottom (ISIC 2018, $M=7$):** skin lesion; 89.58% DSC (+0.70%). Gains are relative to the DA-TransUNet values in Tables I–III.

C. Computational Cost and Parameter Profile

Table IV reports parameter counts and FLOPs evaluated with a standardized profiling tool across all architectures. While spatial windowing and low-rank projection reduce the asymptotic complexity of the core affinity matrix (Section III), total computational cost is primarily determined by the hybrid CNN–ViT backbone ($\sim 85\text{M}$ parameters). Furthermore, by preserving full channel dimensions C (omitting the $C/16$ bottleneck compression) and placing an additional attention block at the fine 112×112 skip resolution, ApproxDABlocks emphasize representation fidelity. ApproxDA-TransUNet is thus conceived as a controlled diagnostic vehicle for probing context dynamics and approximation mechanisms rather than a specialized low-latency edge architecture.

TABLE IV PARAMETERS AND FLOPs (FVCORE, ONE 224×224 SLICE)

Method	Params (M)	GFLOPs	Attn. GFLOPs
DA-TransUNet [2]	106.56	30.2	0.87
ApproxDA (gate=pam, $M=7$)	108.98	32.1	2.73
ApproxDA (gate=learn, $M=7$)	109.90	32.1	2.74

One multiply-add is counted as one FLOP. Parameter counts and FLOPs reflect active forward-pass modules at 224×224 input resolution (three attention blocks in DA-TransUNet, four in ApproxDA-TransUNet).

V. ABLATION STUDIES AND ANALYSIS

This section investigates how ApproxDA-TransUNet behaves across approximation parameters and segmentation tasks through systematic ablation studies.

A. Window Size and Rank Ablation

We first examine how spatial window size and projection rank govern segmentation performance. Table V evaluates the window size $M \in \{7, 28, 56, 112\}$ under fixed gate=pam, $r=32$, and $G=8$ on Synapse, alongside rank scaling at $M=112$. Note that setting $M=112$ establishes global spatial scope per feature map while maintaining low-rank key/value projection ($r=32$).

TABLE V WINDOW SIZE AND RANK ABLATION ON SYNAPSE (GATE=PAM, $G=8$)

Config	M	r	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
Windowed + low-rank	7	32	78.64	31.09
Windowed + low-rank	28	32	80.94	<u>27.49</u>
Windowed + low-rank	56	32	78.90	35.23
Global + low-rank	112	32	<u>79.44</u>	27.26
Global + low-rank	112	64	78.93	31.21

Upper block: only M varies. Lower row: rank variation relative to $M=112, r=32$. $M=112$ establishes map-wide spatial scope with rank- r projection.

The Synapse DSC-vs- M trajectory exhibits clear *non-monotonicity*: it reaches peak performance at an intermediate scale of $M=28$ (80.94%), while highly localized partitioning ($M=7$: 78.64%), larger windows ($M=56$: 78.90%), and map-wide attention ($M=112$: 79.44%) all yield lower accuracy. The

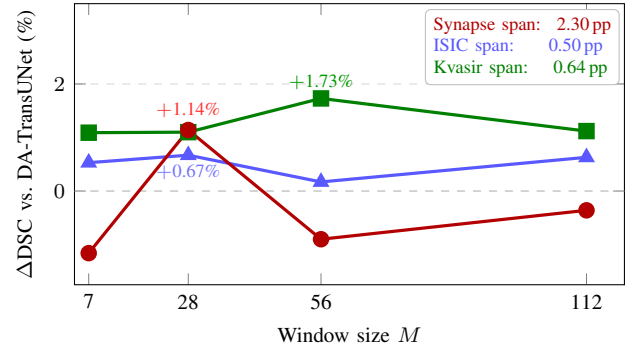


Fig. 4. ΔDSC vs. window size M (gate=pam, $r=32$) across three segmentation tasks, relative to the DA-TransUNet values in Tables I–III. Curve colors: Synapse (red), ISIC 2018 (blue), Kvasir-SEG (green). Per-dataset DSC spans in the annotation box (top right).

2.30 pp DSC spread highlights pronounced window sensitivity on multi-organ CT. Furthermore, scaling rank from $r=32$ to $r=64$ at $M=112$ yields no performance gain (78.93% vs. 79.44%), indicating that spatial receptive scope exerts a substantially stronger influence on segmentation accuracy than projection rank within this operating regime.

B. Cross-task Ablation and Global Context Sensitivity

We next assess whether attention approximation responds consistently across distinct anatomical segmentation tasks. Although ApproxDA-TransUNet employs identical approximation mechanisms across benchmarks, its empirical sensitivity to window selection diverges significantly. On Synapse multi-organ CT, the DSC-vs- M curve (Figure 4) exhibits a sharp peak at $M=28$ spanning a 2.30 pp range. Conversely, on Kvasir-SEG, the curve is remarkably flat (0.64 pp spread) with an optimum at $M=56$; on ISIC 2018, the performance range across $M \in \{7, 28, 56, 112\}$ is only 0.50 pp. The exact raw DSC values across $M=7/28/56/112$ (gate=pam, $r=32$) are 89.53/89.54/90.17/89.56% on Kvasir-SEG and 89.41/89.55/89.05/89.51% on ISIC 2018. Across all three datasets, optimal windowing outperforms the baseline DA-TransUNet metrics in Tables I–III by +1.14% (Synapse, $M=28$), +1.73% (Kvasir-SEG, $M=56$), and +0.67% (ISIC 2018, $M=28$).

We formalize this cross-task behavior through the empirical metric **Global Context Sensitivity (GCS)**, defined as the performance span over the window sweep:

$$\Delta\text{DSC} = \max_M \text{DSC}(M) - \min_M \text{DSC}(M), \quad (7)$$

measured under uniform gate=pam, $r=32$. Empirically, $\Delta\text{DSC} = 2.30\text{ pp}$ on Synapse, 0.64 pp on Kvasir-SEG, and 0.50 pp on ISIC 2018—reflecting a $3.6\times$ and $4.6\times$ ratio relative to polyp and lesion segmentation, respectively. Rather than serving as an a priori predictive descriptor, GCS provides a quantified diagnostic of how strongly model accuracy is coupled to spatial context scale. High GCS reflects a sharp optimization landscape requiring careful window tuning, whereas low GCS demonstrates broad stability across receptive scopes.

Importantly, high GCS does not indicate that approximation degrades performance, but rather characterizes performance sensitivity to window calibration.

These differences align with the contrasting anatomical context requirements of the tasks. Multi-organ abdominal CT segmentation involves complex inter-organ spatial dependencies, where both overly localized and overly broad receptive fields prove suboptimal. In contrast, polyp and skin lesion segmentation are heavily driven by local texture, color contrast, and boundary continuity, where windowed attention reliably preserves essential discriminatory cues.

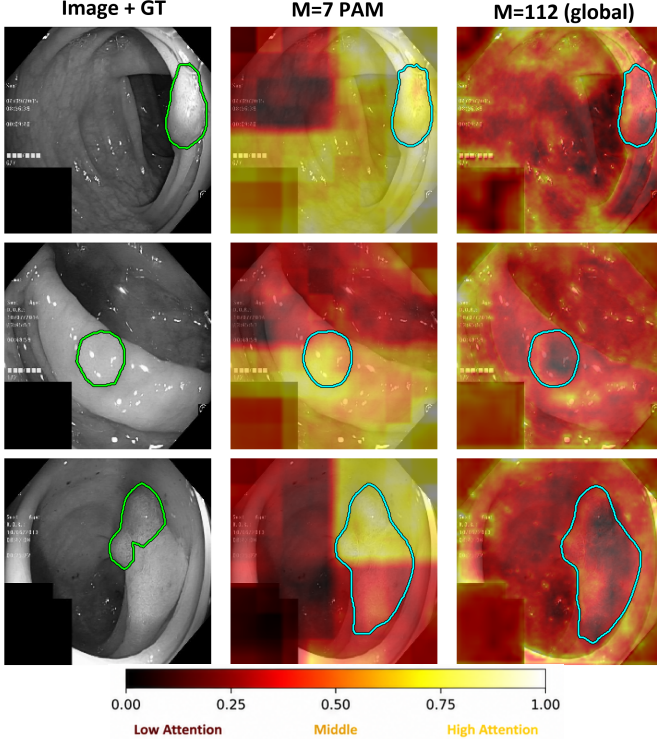


Fig. 5. PAM contribution maps ($\|output - input\|_2$, per-panel normalised; colorbar below) on three Kvasir-SEG polyp images. Each row: input with GT contour (left), windowed PAM ($M=7$; center), map-wide PAM ($M=112$; right). Brighter color indicates larger PAM contribution; panels are independently normalised.

Attention contribution maps corroborate this interpretation. Comparing windowed attention ($M=7$) with map-wide attention ($M=112$) on Kvasir-SEG polyp images, windowed attention concentrates 62.3% of its total activation mass on the polyp target compared to 34.2% for map-wide attention, exhibiting higher concentration in 9 of 10 evaluated samples (Figure 5). Localized windowing naturally focuses attention along lesion boundaries while suppressing diffuse background responses.

C. Routing Mechanism Ablation

We analyze branch routing by evaluating PAM-only, CAM-only, and learnable gating on Synapse under fixed $M=7, r=32, G=8$ (Table VI).

TABLE VI GATE MODE ABLATION ON SYNAPSE ($M=7, r=32, G=8$)

Gate Mode	g	DSC (%) $\uparrow$	HD95 (mm) $\downarrow$
gate=pam	1	78.64	31.09
gate=cam	0	78.26	30.59
gate=learn	$\approx 0.5 \cdot 1^*$	77.78	34.29

*Learned per-channel gate values converged near 0.5 (Figure 6).

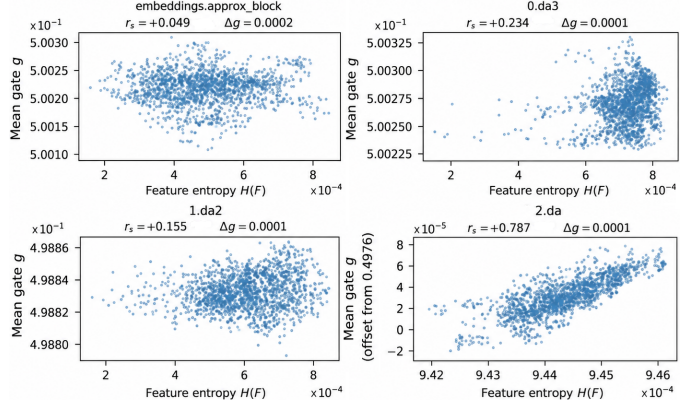


Fig. 6. Mean gate value g vs. feature entropy $H(F)$ across Approx-DABlock locations ($M=7$, gate=learn): bottleneck, 28^2 , 56^2 , and 112^2 skips. Across all locations, the mean gate exhibits minimal sample variance ($\Delta g \leq 0.0002$) around 0.5. Spearman rank correlations with entropy are $r_s = 0.05, 0.23, 0.16, 0.79$, respectively.

PAM-only routing attains the highest DSC (78.64%), closely followed by CAM-only (78.26%) and dynamic gating (77.78%). In our Synapse experiments, the learnable gate converged to a near-uniform state ($g \approx 0.5$): Figure 6 shows that across all four block stages, mean gate values remain within $\Delta g \leq 2 \times 10^{-4}$ of location-specific values near 0.5. Although gate values correlate with feature entropy at certain stages (up to $r_s=0.79$ at the 112^2 skip), the absolute modulation range remains narrow across samples. A supplementary run with $M=14, r=64$ (78.04% DSC) revealed consistent near-uniform routing. We document this as an empirical phenomenon in multi-organ CT training; exploring specialized regularization and alternative gating topologies presents an interesting direction for future work. Consequently, PAM-only routing serves as the standard configuration for cross-task window ablations.

VI. CONCLUSION

This paper presented ApproxDA-TransUNet, a dual-attention approximation framework for medical image segmentation featuring independently controllable window size, projection rank, and channel grouping. Across three representative clinical benchmarks, ApproxDA-TransUNet achieves consistent Dice gains over DA-TransUNet with appropriate window configurations, demonstrating that attention approximation directly modulates representational capacity and accuracy rather than acting solely as an efficiency mechanism.

Our systematic ablations further reveal that context sensitivity diverges substantially across clinical targets, a property

we formalize via empirical Global Context Sensitivity (GCS): multi-organ CT segmentation requires precise scale tuning, whereas polyp and skin lesion targets exhibit broad robustness across receptive window scales. In addition, our empirical findings highlight that learnable dual-branch routing tends toward static uniform blending on multi-organ CT without surpassing single-branch PAM routing.

Limitations and Future Work. In accordance with benchmark literature conventions, reported metrics reflect single-run performance on standard splits, and certain baseline figures are cited from published studies under varying training protocols. Furthermore, while GCS effectively summarizes empirical context sensitivity across the evaluated benchmarks, developing predictive context heuristics prior to exhaustive sweeping and validating them across broader anatomical modalities and architectures remain valuable directions for future study. Finally, translating theoretical attention-matrix reductions into end-to-end inference acceleration via specialized kernel implementations and lightweight backbones represents a natural next step. We hope this work encourages principled, task-aware exploration of attention approximation across medical imaging domains.

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